

Effects of yoga on sleep quality among the geriatric population: Systematic review and meta-analysis[☆]

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ABSTRACT

Background: Sleep disorders are prevalent in the elderly and tend to have a substantial influence on their daytime functioning as well as their quality of life. Pursuant to the latest studies, yoga may have benefits for enhancing a variety of aspects of sleep in this demographic. This meta-analysis and systematic review aim to give an up-to-date and complete assessment of the effects of yoga on sleep quality in the elderly.

Methods: We conducted a thorough search of electronic databases (MEDLINE/PubMed, Cochrane Library, Scopus, and Science Direct) for English-language articles. We focused on randomized control trials and experimental studies; for evaluating yoga interventions aimed at improving sleep quality in the elderly. After assessing the risk of bias, we meta-analyzed the extracted data using a random-effects model to assess various sleep outcome parameters.

Results: Five studies with a total of 467 participants that met our inclusion criteria were included in this study. The meta-analysis demonstrated that yoga has a significant positive effect on sleep quality. Specifically, yoga interventions were associated with improvements in sleep disturbance (standardized mean difference [SMD]: -0.63 ; 95 % CI -1.01 to -0.25), sleep duration (SMD: -0.96 ; 95 % CI -1.89 to -0.02), sleep efficiency (SMD: -0.63 ; 95 % CI -1.01 to -0.25), and the PSQI-Global score of sleep quality (SMD: -0.96 ; 95 % CI -1.63 to -0.28). These results were statistically significant.

Conclusions: Our meta-analysis indicates that yoga practices can lead to improvements in sleep quality among the elderly, demonstrating meaningful progress based on the results of conducted studies. To enhance the significance and generalizability of these findings, future studies should consider larger sample sizes and a higher number of RCTs, which can contribute to minimizing bias.

Background

Ageing is an irreversible process marked by physiological changes that might affect health. The growing number of elderly people will result in significant variety of medical difficulties, including sleep disturbances (Shree Ganesh et al., 2021). Ageing is a global phenomenon, but there wasn't enough information available on a regional level. Additionally, The American Academy of Sleep Medicine's International Classification of Sleep Disorders, Third Edition, states that while insomnia is very common in the senior population, reports from various

regions of the world vary considerably (Gulia and Kumar, 2018). This sleep disturbance has a negative impact on the elderly's daytime functioning and lowers their quality of life (Hariprasad et al., 2013). Lack of sleep can entail enormous socioeconomic and health consequences, appearing as morbidity, cognitive impairment, and diminished quality of life (Panjwani et al., 2021; Bandyopadhyay et al., 2023) and also insufficient physical activity, poor sleep hygiene, and excessive daytime napping are all variables that contribute to sleep impairment in older persons. Despite their importance in healthcare and the economy, sleep-related disorders in the elderly are frequently ignored and

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misattributed to the aging process. Pharmacological therapies for sleep difficulties, such as benzodiazepines, are not without side effects, particularly in the elderly population. Adverse effects of such drugs include dependence, cognitive impairment, and rebound sleeplessness. (Bankar et al., 2013).

Recent data reveals that elderly people are drawn to simple physical activities that boost their well-being and inner happiness, which ultimately contributes to their longevity. (Choudhary et al., 2019). Yoga has achieved a global reputation as a non-pharmacological solution due to its comprehensive approach. It improves cardiovascular and metabolic health, mood, sleep quality, body balance, and mobility (Choudhary et al., 2019; Wu et al., 2015). The yoga's psychological and physical benefits appeal with the elderly, potentially relieving age-related health and emotional concerns. (Patel et al., 2012).

Yoga is increasingly being investigated for its therapeutic benefits in older persons (Hariprasad et al., 2013) which is linked to psychosocial benefits via the management of aging-related health and emotional challenges (Patel et al., 2012). Recent research demonstrates its effectiveness in enhancing sleep characteristics in the elderly. (Chen et al., 2010). Yoga has showed potential in reducing sleep onset delay,

increasing sleep duration, and improving overall sleep quality (Mooventhan & Nivethitha, 2017). Furthermore, mindfulness, a component of yoga, promotes better sleep by moderating stress-related physiological changes and boosting melatonin levels. (Wang et al., 2020).

Despite the well-established link between sleep quality and general well-being, there has been no comprehensive systematic review or meta-analysis of randomized controlled trials or the trials with control group investigating the efficacy of yoga in improving sleep quality in the elderly. As a result, the main goal of this study is to fill this gap by analyzing the available literature in order to evaluate the impact of yoga therapies on sleep quality in the geriatric population.

Methods

The current systematic review and meta-analysis is reported in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) guidelines (Page et al., 2021) (Fig. 1).

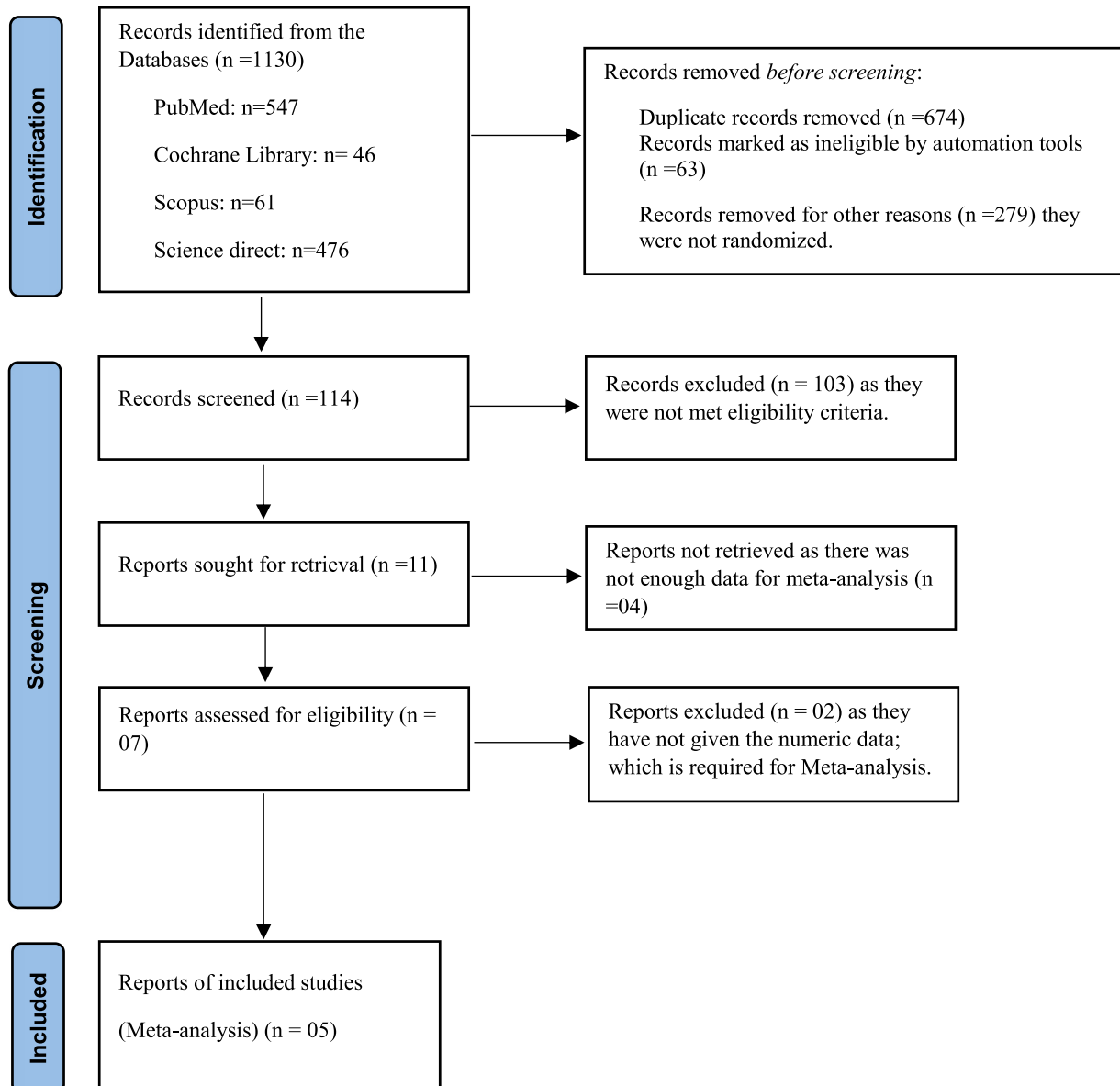


Fig. 1. A PRISMA diagram illustrating the search strategy for the review.

Search Strategy

We searched on electronic databases (MEDLINE/PubMed, Cochrane Library, Scopus, and Science Direct) for articles published in English from January 2000 to December 2022; where the relevant articles found and also to update the recent 20 years of evidences rather than the studies from inception. The search strategy combined terms related to or describing a combination of search terms which are same in all the databases, such as yoga OR hatha yoga OR asana OR pranayama OR meditation OR OM meditation OR relaxation AND old age OR elderly OR older adults OR geriatric AND sleep OR insomnia OR sleep deprivations. Published reviews and the reference lists of retrieved publications were searched manually.

Eligibility criteria

Research trials were included in the review if it satisfies the following criteria:

Included studies were randomized controlled trials (RCTs) and also experimental studies done with randomization comparing yoga practices to any other control treatment. To be included, studies had to report on older adults (age above 60 years) with lower sleep quality. Studies involving practices like Thai yoga or mimicking like yoga practices were excluded. Included the Yoga practices such as asana, pranayama, meditation and relaxation and the control group did not given with any intervention during the study period, and the sleep quality assessed using Pittsburgh sleep quality index (PSQI questionnaire) (Buysse et al., 1989) among geriatric population.

Data extraction and reliability

The titles and abstracts of all the records are screened for relevance and full texts papers were only selected and reviewed by (SP and MK) two independent authors, then the data were extracted using an Excel (Microsoft) Spreadsheet. The risk of bias of included studies were also assessed. Data was extracted primarily based on study information (E.g. First author, Year of publication, study period), study design (RCT & experimental study, randomized and included control group), participant characteristics (E.g. Age, country), Yoga intervention characteristics (E.g. Intervention type of practice, Duration), Control intervention characteristics (E.g. Active control), outcome measures (Primary and secondary outcome measures of sleep quality among elderly) from each study.

Study outcomes

Based on the availability of sufficient data among the validated questionnaires for the assessment of the sleep quality among the elderly in various studies, the PSQI sleep assessing tool is selected and included as the main outcome for the meta-analysis. This questionnaire covers various aspects of sleep, including subjective sleep quality, sleep latency, sleep duration, sleep efficiency, sleep disturbances, medication use related to sleep, and daytime dysfunction. These components collectively provide a comprehensive evaluation of sleep quality.

Quality assessment

A bias analysis tool from the Cochrane risk of bias tool was used to evaluate the caliber of the included in the study. Two researchers (SP and MK) independently evaluated the listed studies level of quality. Based on the production of random sequences, allocation concealment, blinding of participants or outcome assessors, insufficient outcome data, selective result reporting, and other forms of bias, each study is categorized as low risk, high risk, or unclear risk.

Data synthesis and analysis

Meta-analyses were carried out utilizing R statistical software with metafor package. To quantify the effect size, the Standardized Mean Difference (SMD) along with a 95 % Confidence Interval (CI) was computed. In order to evaluate the level of heterogeneity among the included studies, statistical indicators such as the I^2 statistic and the Cochran Q test were employed.

Results

Literature search

We have identified total of 1130 publications through electronic data bases, of which 1016 were duplicates are removed, they were excluded as different databases used for searching articles and they were not meeting the eligibility criteria (non-randomization, mixed population including old age, not given data). Further publications were identified through reference lists or other sources. Title and abstracts of the 114 potential publications were screened, only 07 articles met the full text appraisal, of which 05 articles met all inclusion criteria and included for the Meta analysis (Table 1).

Study characteristics

The Characteristics of the studies included were in the Table 1. Five studies met the inclusion criteria, four studies were RCTs (Baklouti et al., 2023; Cheung et al., 2014; Hariprasad et al., 2013; Shree Ganesh et al., 2021) and one was a quasi-experimental study with randomization (Chen et al., 2010). Out of five studies, two studies were conducted in India (Hariprasad et al., 2013; Shree Ganesh et al., 2021), one study in Tunisia (Baklouti et al., 2023), one study in Taiwan (Chen et al., 2010), and one study in USA (Cheung et al., 2014). Four studies included both sexes as subjects (Baklouti et al., 2023; Chen et al., 2010; Hariprasad et al., 2013; Shree Ganesh et al., 2021) and one study included only female subjects (Cheung et al., 2014), all five studies included age group of above 60 years. Total number of participants was 467 in which included 224 in the intervention group and 228 in the control group (Table 1). Selected trials used PSQI Questionnaire to assess the sleep quality.

Bias assessment

Cochrane Risk of Bias Tool (RoB 2; Version-2) (Table 2) was used to assess the risk of bias of the included studies. Out of the five included studies, four (Chen et al., 2010; Cheung et al., 2014; Hariprasad et al., 2013; Shree Ganesh et al., 2021) were rated as low risk for random sequencing. Regarding allocation concealment, four studies (Chen et al., 2010; Cheung et al., 2014; Hariprasad et al., 2013; Shree Ganesh et al., 2021) were rated as low risk, while one study (Baklouti et al., 2023) it was unclear. Blinding of participants and personnel reporting two studies (Cheung et al., 2014; Hariprasad et al., 2013) were low risk and rest of the three studies (Baklouti et al., 2023; Chen et al., 2010; Shree Ganesh et al., 2021) were rated high risk. The blinding outcome assessment rated as two (Hariprasad et al., 2013; Shree Ganesh et al., 2021) were unclear, two (Baklouti et al., 2023; Chen et al., 2010) as high risk and one (Cheung et al., 2014) as low risk. Three studies (Cheung et al., 2014; Hariprasad et al., 2013; Shree Ganesh et al., 2021) were rated as Low risk in data selective reporting and other two (Baklouti et al., 2023; Chen et al., 2010) were unclear. Two studies (Cheung et al., 2014; Shree Ganesh et al., 2021) on incomplete outcome data were rated as low risk and other three (Baklouti et al., 2023; Chen et al., 2010; Hariprasad et al., 2013) were unclear. No details regarding the additional bias sources that are considered to be high risk are provided in any of the studies (Fig. 2 & Fig. 3).

Table 1
Study and sample characteristics.

S. no	Author	Country	Age	Sex	Sample size	Study design	Intervention for the experimental group	Intervention for control group intervention details
1.	Shree Ganesh, H. R., et al. (2021)	INDIA	Between 60 and 75 years of age	Both	T= 96 E=48 C=33	A randomized controlled trial	Yoga protocol followed as 3 sessions per week for 3 months	Wait-list group No practices
2.	Hariprasad, V. R., et al. (2013)	INDIA	Above 60 years	Both	T=120 E=62 C=58	A randomized controlled trial	A Validated yoga program for 60 mints daily for 1 month; Following these participants were asked to practice the same daily until the end of 6 months. in the 2nd and 3rd months of the study, they received 1-h supervisory sessions	Did not receive any intervention during the study period
3.	Chen, K. M., et al. (2010)	TAIWAN	Elders aged 65 years or older	Both	T=55 E=31 C=24	Quasi-experimental study with Randomization	The silver yoga exercise program was implemented for.70 min per session; 3 times per week for 24 weeks	Control group participants were instructed to follow their usual daily activities in the institution
4.	Baklouti, S., et al. (2022).	TUNISIA	Age above 65 years	Both	T= 160 E=65 C=95	A randomized controlled trial	The experimental group had Hatha yoga sessions of 80 min twice a week for 8 weeks.	The control group received no treatment.
5.	Cheung, C., et al. (2014)	USA	Between 65 and 90 years	only women	T=36 E=18 C=18	A pilot randomized controlled trial	One 60 min of yoga session per week for 8-weeks included hatha yoga intervention.	received the same yoga intervention after the treatment group completed the program at the end of 8 weeks

Abbreviation used: T= Total sample size; E=Experimental group; C=Control group;

Table 2
Risk of Bias assessment tool (RoB 2).

S. no.		Random sequence generation	Allocation concealment	Selective reporting	Other sources of bias	Blinding (participants and personnel)	Blinding (outcome assessment)	Incomplete outcome data
1	Shree Ganesh, H. R., et al. (2021)	Low risk	Low risk	Low risk	High	High	Unclear	Low risk
2	Hariprasad, V. R., et al. (2013)	Low risk	Low risk	Low risk	High	Low risk	Unclear	Unclear
3	Chen, K. M., et al. (2010)	Low risk	Low risk	Unclear	High	High	High	Unclear
4	Baklouti, S., et al. (2022)	Unclear	Unclear	Unclear	High	High	High	Unclear
5	Cheung, C., et al. (2014)	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk	Low risk

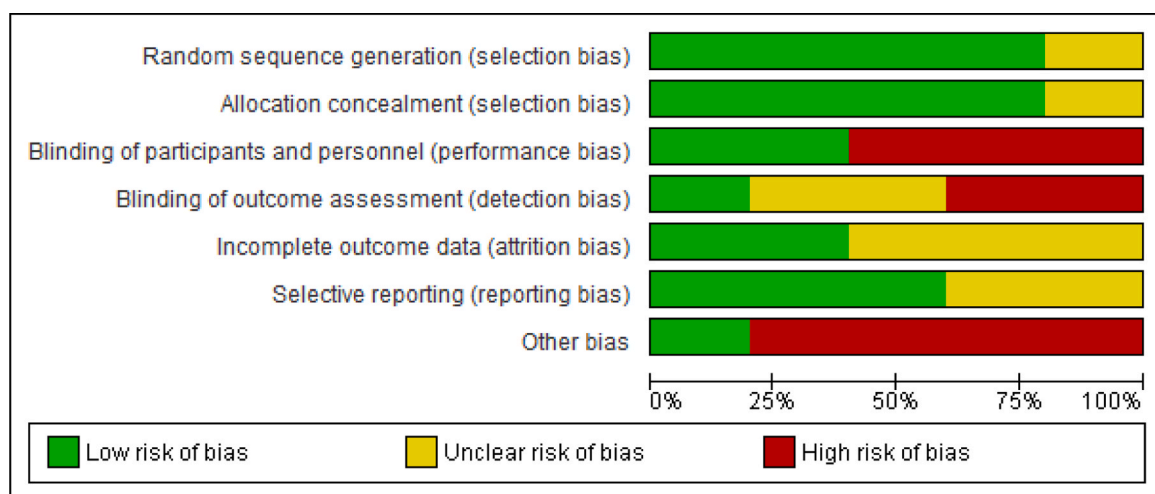


Fig. 2. Risk of bias summary.

Meta-analysis

In the included studies, the quality of sleep was evaluated using Pittsburgh Sleep Quality Index (PSQI). Meta-analysis was done using the

availability of data with the following components from PSQI questionnaire; sleep disturbance, sleep duration, sleep efficiency and the global score.

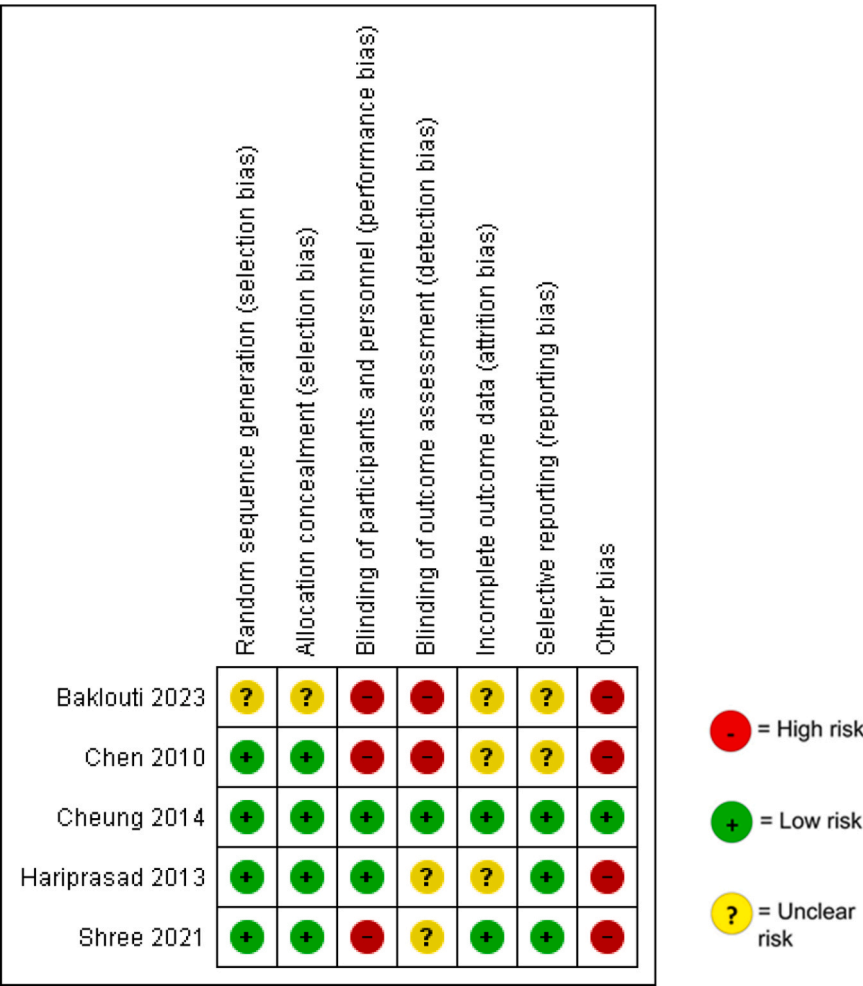


Fig. 3. Risk of Bias domains judgment.

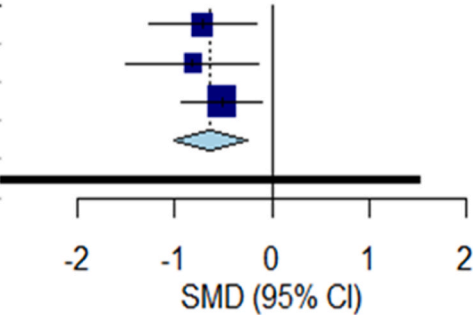
Sleep disturbance (SDe)

Fig. 4 displays the forest plot of 3 studies (Chen et al., 2010; Cheung et al., 2014; Shree Ganesh et al., 2021) reporting the sleep disturbance among the included studies. The pooled effect of yoga intervention reveals a significant improvement of sleep disturbance with the moderate effect size (SMD: -0.63; 95 % CI -1.01; -0.25) and low Heterogeneity ($I^2 = 0\%$).

Sleep duration (SD)

Fig. 5 shows the forest plot of 3 studies (Chen et al., 2010; Cheung et al., 2014; Shree Ganesh et al., 2021) reporting the sleep duration among the included studies. The pooled effect of yoga intervention reveals a significant improvement of sleep duration with the strong effect size (SMD -0.96 with 95 % CI -1.89; -0.02) and the considerable Heterogeneity ($I^2=31\%$).

Source	SMD (95% CI)
Chen, K. M., et al. (2010)	-0.72 [-1.27; -0.17]
Cheung, C., et al. (2014)	-0.81 [-1.50; -0.13]
Shree Ganesh, H. R., et al. (2021)	-0.52 [-0.92; -0.11]
Total	-0.63 [-1.01; -0.25]
95% PI	[-2.78; 1.52]



Heterogeneity: $\chi^2_2 = 0.68$ ($P = .71$), $I^2 = 0\%$

Fig. 4. Forest plot chart for sleep disturbance.

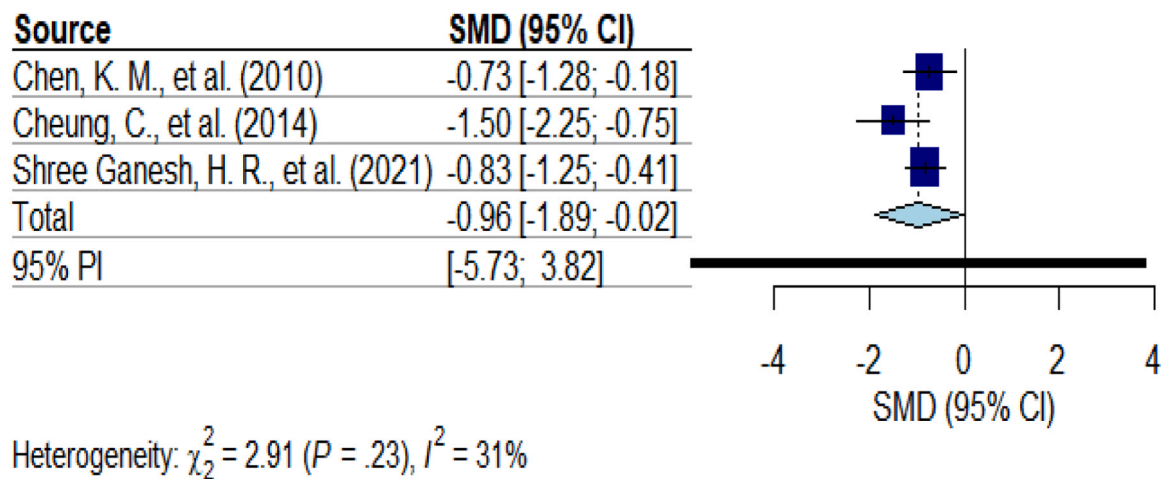


Fig. 5. Forest plot chart for sleep duration.

Sleep efficiency (SE)

Three studies have reported the sleep efficiency outcomes (Chen et al., 2010; Cheung et al., 2014; Shree Ganesh et al., 2021). Result presented as forest plot shown in Fig. 6, it was statistically significant (SMD: -0.63, 95 % CI -1.01; -0.25) with less Heterogeneity ($I^2=0$ %).

PSQI – Global score (GS)

The global score of PSQI questionnaire was recorded in five studies (Baklouti et al., 2023; Chen et al., 2010; Cheung et al., 2014; Hariprasad et al., 2013; Shree Ganesh et al., 2021) were analyzed to assess the effect of Yoga intervention on the overall quality of sleep among the elderly population. Result presented as forest plot shown in Fig. 7, it was statistically significant in improving the sleep quality (total: -0.96, 95 % CI -1.63; -0.28) compared with control group which has considerable Heterogeneity ($I^2= 64$ %).

Discussion

In this study, we have found that yoga intervention has significant impact on sleep quality among the geriatric population. There was no systematic review and meta-analysis available that explicitly focused on yoga for improving sleep quality for geriatrics. Ours is the first systematic review and meta-analysis with 5 studies with 467 participants that to focus on the effects of yoga on sleep quality for older adults. Findings suggest that moderate sized beneficial effects can be achieved for sleep duration, sleep disturbance, and sleep efficiency and PSQI

global score for the older adults with yoga intervention. While other systematic reviews have included on studies that recruited older adults with clinical conditions, this review is the first to provide a comprehensive overview of the effects of yoga on sleep quality in an older adult population not characterized by a specific disease. A recent meta-analysis demonstrated a moderate effect size, ($g = 0.47$, 90 % CI [0.24, 0.70], $p = 0.0007$), suggesting that the mind body medicine (MBM) intervention had a positive impact on sleep quality (Weber et al., 2020). Another meta-analysis of 22 RCTs indicates that yoga interventions have a significant positive effect on various aspects of physical function and quality of life. Specifically, for sleep quality, yoga demonstrates a substantial and favorable effect with an effect size (ES) of 0.65 (95 % CI 0.41–0.88, $p < 0.001$) when compared to inactive controls (Sivaramakrishnan et al., 2019). All the five studies included for meta-analysis were having nearly similar interventions as they included warm up (joint exercise), asana (postures), pranayama (breathing practice), and meditation practices. The duration of interventions of the included studies ranges from 8 weeks to 24 weeks.

A recent systematic review (Bandyopadhyay et al., 2023) used seven studies (four RCTs, one pre-posttest study, one longitudinal, and one cross-sectional study) reported that practicing yoga with various postures and stretching exercises improves joint flexibility, muscular strength, and blood circulation and prevents cartilage dystrophy, thereby maintaining sleep quality. Furthermore, yoga activities can lower nerve tension, balance the sympathetic nervous system's excitement suppression process, alleviate weariness, improve sleep and deep sleep, and stimulate endorphin release and oxygen saturation, all of which contribute to better sleep quality (Bandyopadhyay et al., 2023).

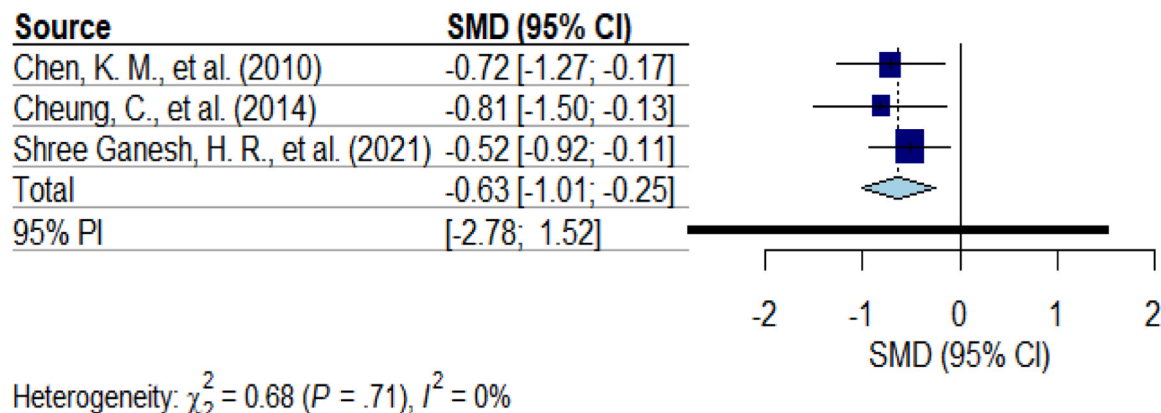
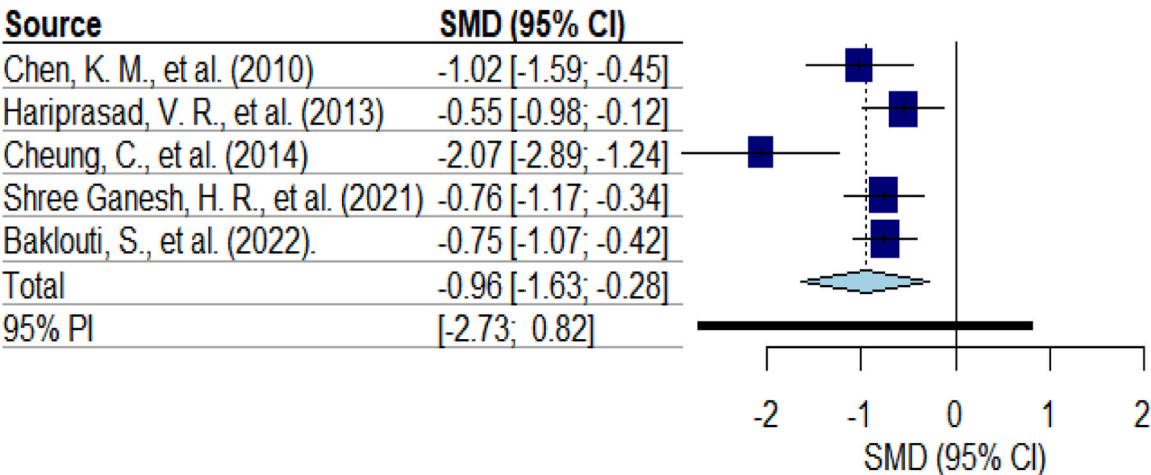


Fig. 6. Forest plot chart for Sleep efficiency.



Heterogeneity: $\chi^2_4 = 11.06$ ($P = .03$), $I^2 = 64\%$

Fig. 7. Forest plot chart for PSQI – Global score.

Participation in a regular yoga session lowers the levels of norepinephrine and cortisol in the human body and mind, providing relaxation for the body and so reducing daytime dysfunction and improving sleep quality. Yoga positions have been associated to greater levels of the neurotransmitter gamma-amino butyric acid (GABA), as measured by nuclear magnetic resonance spectroscopy (NMRs), which would enhance cortical inhibition and, as a result, aid in sleep. Guided relaxation may promote sleep since it is associated with greater relaxation than passively lying in bed. In healthy populations, pranayama and meditation help to enhance melatonin levels, which promote better sleep quality. As a result, it is clear that participating in a yoga program of at least three sessions per week can significantly benefit all aspects of good sleep organization, including a reduction in sleep disturbances, sleep latency, time taken to fall asleep, and an increase in sleep duration and sleep efficiency, all of which can improve sleep quality (Bandyopadhyay et al., 2023). Sleep disruptions in the elderly can have a considerable detrimental impact on daily function, including morbidity, poor cognitive performance, and lower quality of life (Bandyopadhyay et al., 2023; Hariprasad et al., 2013). Furthermore, it can cause increased daytime sleepiness, restlessness, and significant respiratory, depression, and anxiety symptoms (Shree Ganesh et al., 2021). Insomnia is also a risk factor for accidents and falls, which are the leading causes of death in older persons, and is thus related with greater morbidity and mortality rates in older populations. Mindfulness, as a key component of yoga, reduces sleep disruption by boosting melatonin levels, decreasing hyper arousal, and resolving stress-related cardiac and respiratory problems (Wang et al., 2020). As a result, daily yoga practice has been reported to be a safe intervention in enhancing several elements of sleep, quality of sleep, and quality of life. In the aged, higher sleep quality is connected with better health, reduced daytime sleepiness, greater well-being, and better psychological functioning (Hariprasad et al., 2013).

Our study has some limitations. Firstly, the studies we reviewed included older populations with minor comorbidities which may not be comparable in population without comorbidities. Consequently, the applicability of our findings and recommendations to the broader older adult population may be limited, despite the statistical significance of our results. The diverse nature of these populations means that individuals exhibited different pathophysiological behaviors, a wide range of severity levels, and the presence of comorbidities, all of which could potentially influence the magnitude of the reported results.

Conclusions

In conclusion, our systematic review and meta-analysis indicate that yoga interventions may have a positive and statistically significant impact on improving sleep quality among the geriatric population. Further research and tailored yoga interventions are warranted to optimize sleep-related benefits for the geriatric population. Integrating yoga into geriatric care strategies could lead to enhanced sleep quality and improved quality of life for older individuals.

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CRediT authorship contribution statement

S. Prashanth: Conceptualization, Data curation, Writing - original draft. **V. Nalini:** Supervision, Validation. **K. Maheshkumar:** Formal analysis, Methodology, Supervision, Validation, Writing - review & editing. **J. Thanalakshmi:** Supervision, Wrting - review & editing. **S. Priyanka:** Data curation, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. There have no known interests that influence the work reported in this paper.

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